Conway's Law for AI Cheat Sheet

Who should own which part of your AI system, and the team shapes that break it. One page.

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1 · THE LAW

> *"Any organization that designs a system (defined broadly) will produce a design whose structure is a copy of the organization's communication structure." Melvin Conway, 1968*

- **In AI:** every owner of a piece (model access, prompts, retrieval, evals, guardrails, tracing) becomes a seam in the system.
- **Inverse Conway maneuver:** pick the target architecture first, then design teams that would naturally build it.
- **Evidence:** org-structure metrics predicted failing Windows Vista binaries better than code metrics (Microsoft Research, 2008).

2 · FOUR TEAM TYPES, FOR AI

Stream-aligned (product)	Owns the AI feature: prompts, tools, UX, its eval cases
Platform	Gateway + routing, eval harness + CI gate, tracing, cost, guardrail enforcement, as a self-service
Complicated subsystem	Retrieval/ranking at scale, fine-tuning, custom models
Enabling	Coaches teams on prompts, evals, agent patterns, AI security, then steps away

3 · WHO OWNS WHAT

Component	Owner
Model keys, access, routing	AI platform
Prompts, tool definitions	Product team
Eval harness, CI gate	AI platform
Eval cases ("what good means")	Product team
RAG pipeline	Product team first; specialist team once shared
Guardrail policy	Security / risk, via enabling team
Guardrail enforcement	AI platform, in the gateway
Tracing, cost dashboards	AI platform
Fine-tuning, custom models	Complicated-subsystem team

4 · INTERACTION MODES

Collaboration	First agent or RAG feature, while both teams learn. Set an end date.
X-as-a-Service	Steady state for gateway, evals, tracing, shared retrieval. No tickets needed.
Facilitating	Enabling team teaches eval writing, guardrails, agent patterns.

> **Most common failure:** the platform team stays in collaboration with everyone, so it never becomes a service.

5 · WARNING SIGNS

- Every AI feature waits in one central team's backlog.
- Each team runs its own gateway, vector store and eval script.
- The AI CoE writes guidelines nothing enforces.
- Prompt or model changes ship with no eval gate.
- You can't trace one request across two teams' services.
- Agents have access everywhere, so they couple things no team agreed to couple.

6 · SETUP BY COMPANY SIZE

Under ~30 engineers	No AI team. Product teams use a model API; one shared wrapper and eval script.
~30–150 engineers	Product teams + a thin AI platform team (2–4 people) for gateway, evals, tracing.
150+ engineers	Platform grouping, specialist teams for retrieval/models, an enabling team. "Hub and spoke."

7 · WHAT AN AI PLATFORM SHOULD PUBLISH

- Models offered, with cost per call and latency targets
- How to get access: self-service keys, no ticket
- How to run evals and read results; the CI gate rules
- Tracing format and where to find a request's trace
- Guardrails enforced by default, and how to request exceptions
- Owner, on-call and support channel

Full guide: Conway's Law for AI · AI agent observability · AI guardrails

Team types: Team Topologies, Skelton & Pais.